A SMOOTH MORPHISM ADMITS LIFTS AFTER AN ETALE COVERING OF A THICKENING OF THE TARGET

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 4713-4733 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 4735-4752, the complete author proof (16 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Smooth morphisms of algebraic spaces, thickenings, etale coverings, and the infinitesimal lifting criterion; the reduction is retained.

Source reference. Original lemma 07K4: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 4752–4752. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 04EW: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 1502–1537. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 05ZH: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 1316–1326. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Consider a commutative diagram*

$$\begin{array}{ccc} X & \longleftarrow & T \\ \downarrow & & \downarrow \\ Y & \longleftarrow & T' \end{array}$$

of algebraic spaces over S where $X \to Y$ is smooth and $T \to T'$ is a thickening. Then there exists an étale covering $\{T'_i \to T'\}$ such that we can find the dotted

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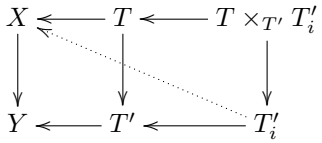

making the diagram commute (for all i).

Proof. Choose an étale covering $\{Y_i \to Y\}$ with each Y_i affine. After replacing T' by the induced étale covering we may assume Y is affine.

Assume Y is affine. Choose an étale covering $\{X_i \to X\}$. This gives rise to an étale covering of T. This étale covering of T comes from an étale covering of T' (by Theorem 05ZH, see discussion in Section 04EW). Hence we may assume X is affine.

Assume X and Y are affine. We can do one more étale covering of T' and assume T' is affine. In this case the lemma follows from Algebra, Lemma 07K4. □